Project Description: Multigraded Homological Algebra and Geometry

Overview. Castelnuovo–Mumford regularity and graded Betti numbers are standard tools linking the geometry of a projective variety to the algebra of its equations. This proposal extends parts of this theory beyond the standard graded setting: to nonstandard $\mathbb{Z}$-gradings, which arise from stacky curves, and to $\mathbb{Z}^r$-gradings, which arise from toric varieties. After reviewing prior support (§1), we develop a Green's-conjecture-type theory of syzygies of stacky canonical curves (§2); structural results and geometric bounds for multigraded regularity (§3); a uniform regularity theory on the Cox category of a toric variety (§4); and open-source software (§5), with broader impacts in §6.

1. Results of Prior NSF Support

From 2020 to 2022, the PI held an NSF Postdoctoral Research Fellowship (MSPRF DMS-2002239, \$150,000), *Asymptotic Syzygies in Algebraic Geometry.* Since research projects and journal timelines extend past the award, the figures below span 2020–2025: 7 arXiv preprints [31, 33, 32, 21, 20, 5, 34] and 7 published articles, including in *Geometry & Topology* [20], *Journal of Algebra* [31], and *Experimental Mathematics* [35]. This work substantially advanced homological algebra on toric varieties (§1.1) and the cohomology of moduli spaces (§1.3).

1.1 Eisenbud–Goto on Products of Projective Spaces The Castelnuovo–Mumford regularity of a projective variety $X \subset \mathbb{P}^r$ measures the complexity of X through the vanishing of its cohomology. Its centrality rests on the fact that a single integer controls cohomology vanishing, the degrees of generators and syzygies, and the eventual linearity of resolutions.

Theorem 1.1. [55] *Let $\mathcal{F}$ be a coherent sheaf on $\mathbb{P}^n$ and $M = \bigoplus_{e\in\mathbb{Z}} H^0(\mathbb{P}^n, \mathcal{F}(e))$ the corresponding section module. The following are equivalent:*

(1) M is d-regular in the sense of [92]*;*
(2) $\beta_{p,q}(M) = 0$ for all $p \geq 0$ and $q > d + p$; and
(3) $M_{\geq d}$ has a d-linear resolution in the sense of [55].

Nearly every use of regularity in the literature—regularity of powers of ideals [48, 81], the Eisenbud–Goto conjecture [90], the construction and boundedness of Hilbert schemes [67, 65], Gröbner basis complexity [8]—passes through these equivalences.

A toric variety X is a well-behaved compactification of a torus and, under mild hypotheses, a quotient $X = (\mathbb{C}^n \setminus \mathbb{V}(B))/(\mathbb{C}^\times)^r$, where B is the *irrelevant ideal* of X; for $X = \mathbb{P}^n$, this is $B = \langle x_0, \ldots, x_n\rangle$. Cox's theorem [46] identifies coherent sheaves on X with B-saturated $\mathbb{Z}^r$-graded modules over the *Cox ring* $S = \mathbb{C}[x_1, \ldots, x_n]$, translating the geometry of X into multigraded commutative algebra. This dictionary is far subtler than in the standard graded case, and its subtleties drive much current work: virtual resolutions and resolutions of the diagonal [11, 72, 26, 60], generalizations of the BGG correspondence [54, 28], and multigraded local cohomology [18, 42].

Extending regularity beyond projective space has a long history. Early developments include Benson's treatment of nonstandard $\mathbb{Z}$-graded rings [9], later developed by Symonds in his groundbreaking work on invariant theory [100, 101], and Hoffman and Wang's theory for products of two projective spaces [73]. In the toric setting, Maclagan and Smith gave a cohomological definition of multigraded regularity [87] that subsumes the weighted and biprojective theories and has become the standard framework for extending the classical theory to toric varieties.

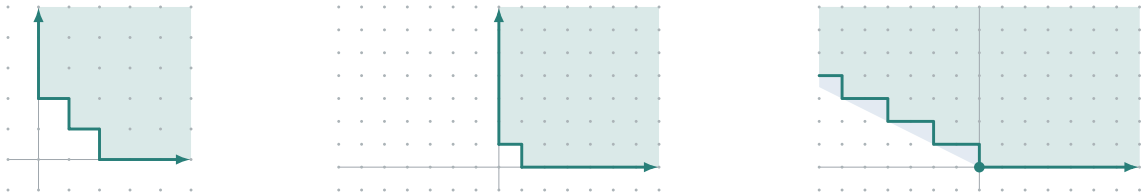
Throughout, X denotes a smooth projective toric variety, so that $\mathrm{Pic}(X) \cong \mathbb{Z}^r$ and the nef cone $\mathrm{Nef}(X)_\mathbb{R} \subset \mathrm{Pic}(X)_\mathbb{R}$ is full-dimensional. We write $\mathrm{Nef}(X) \coloneqq \mathrm{Nef}(X)_\mathbb{R} \cap \mathrm{Pic}(X)$ for the nef monoid and $B_1, \ldots, B_s$ for its unique minimal generators. For an integer vector $\mathbf{v} = (v_1, \ldots, v_t)$ of any length, set $|\mathbf{v}| \coloneqq v_1 + \cdots + v_t$.

Definition 1.2. [87, Def. 6.6] *A coherent sheaf $\mathcal{F}$ on X is* $\mathbf{d}$-regular*, for $\mathbf{d} \in \mathbb{Z}^r$, if, for all $i > 0$,*

$$H^i(X, \mathcal{F}(\mathbf{e})) = 0 \qquad \textit{for all} \qquad \mathbf{e} \in L_i(\mathbf{d}) \coloneqq \bigcup_{\mathbf{v} \in \mathbb{N}^s, |\mathbf{v}| = i} \left(\mathbf{d} - \sum_{j=1}^{s} v_j B_j + \operatorname{Nef}(X) \right).$$

The multigraded regularity *of $\mathcal{F}$ is* $\operatorname{reg}(\mathcal{F}) \coloneqq \{\mathbf{d} \in \mathbb{Z}^r \mid \mathcal{F} \text{ is } \mathbf{d}\text{-regular}\}$.

The regularity $\operatorname{reg}(M)$ of a $\mathbb{Z}^r$-graded S-module M is defined by the analogous vanishing of the local cohomology modules $H^i_B(M)$. Multigraded regularity is thus not a single number but an infinite subset of $\mathbb{Z}^r$, whose boundary can be intricate (Figure 1).



(a) (b) (c)

FIGURE 1. Regularity regions of (a) S/I_Z for three points $Z = \{([1:1],[1:4]),([1:2],[1:5]),([1:3],[1:6])\} \subset \mathbb{P}^1 \times \mathbb{P}^1$; (b) $\mathcal{O}_{\mathcal{H}_2}$; (c) S/I_p for a point $p \in \mathcal{H}_2$.

Multigraded regularity has become a central tool in the study of uniform bounds and Hilbert schemes [88], syzygies and local cohomology [98, 18], virtual resolutions [11, 33], and asymptotic regularity [34]. Yet for more than fifteen years after its introduction, a foundational question remained open: *Is there a multigraded Eisenbud–Goto theorem?* That is, can multigraded regularity be characterized by free resolutions and truncations, as in Theorem 1.1? Direct generalizations fail even for products of projective spaces. Answering a question of Maclagan–Smith, further studied by Botbol–Chardin [18], the PI and coauthors showed that multigraded Betti numbers do not determine regularity [33, Ex. 5.1], ruling out an analogue of condition (2). Likewise, a $\mathbf{d}$-regular module M need not have $M_{\geq \mathbf{d}}$ with a linear resolution, so condition (3) must also be reformulated.

A major advance came with Berkesch, Erman, and Smith's characterization of multigraded regularity on products of projective spaces via virtual resolutions with controlled degrees [11, Thm. 2.9], a counterpart to condition (2). Building on these ideas, the PI and coauthors introduced *quasilinear resolutions* and proved a multigraded Eisenbud–Goto theorem in which quasilinear resolutions of truncations play the role of condition (3).

Theorem 1.3. [33, Thm. A] *Let $\mathcal{F}$ be a coherent sheaf on $X = \mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_r}$ and $M = \bigoplus_{\mathbf{e} \in \mathbb{Z}^r} H^0(X, \mathcal{F}(\mathbf{e}))$ the corresponding section module. For $\mathbf{d} \in \mathbb{Z}^r$, the following are equivalent:*

(1) M is $\mathbf{d}$-regular in the sense of [87]*;*

(2) M has a virtual resolution with degrees as in [11, Thm. 2.9]*; and*

(3) $M_{\geq \mathbf{d}}$ has a $\mathbf{d}$-quasilinear resolution in the sense of [33].

When $r = 1$, Theorem 1.3 recovers the classical Eisenbud–Goto theorem, and (1)⇔(3) holds for every finitely generated M with $H^0_B(M) = 0$. Equally important are its methods: rather than resolving $M_{\geq \mathbf{d}}$ directly, we compute its multigraded Betti numbers via a spectral sequence arising from the Fourier–Mukai transform of $\widetilde{M}$ with Beilinson's resolution of the diagonal [**?**] as kernel. This "resolution of the diagonal" approach is now a central technique [28, 26, 72, 29]. It also yields an effective algorithm, implemented in [3], for testing $\mathbf{d}$-regularity and producing short virtual resolutions.

1.2 Bounds & Asymptotic Behavior of Multigraded Regularity of Ideals Partially order $\operatorname{Pic}(X)$ by $\mathbf{d} \leq \mathbf{e}$ when $\mathbf{e} - \mathbf{d} \in \operatorname{Nef}(X)$. Although nef twists preserve regularity, $\operatorname{reg}(M)$ need not have finitely many minimal elements (pictorially, corners). In [32, Ex. 1.1], the PI, Cranton

Heller, and Sayrafi exhibit a cyclic module on a Hirzebruch surface with infinitely many minimal degrees. We show that for faithful modules such pathologies cannot occur.

Theorem 1.4. [32, Cor. 4.6] *Let M be a finitely generated graded S-module with* $\operatorname{ann}(M) = 0$ *and* $\widetilde{M} \neq 0$. *If its homogeneous generators f_i satisfy* $\mathbf{q} \leq \deg f_i$, *then* $\operatorname{reg}(M) \subseteq \mathbf{q} + \operatorname{Nef}(X)$. *Consequently,* $\operatorname{reg}(M)$ *has finitely many minimal elements.*

The proof uses the fact that $\mathbf{d}$-regularity forces $M_{\geq \mathbf{d}}$ to be generated in degree $\mathbf{d}$; a chamber-and-determinant argument turns a violation of the bound into a nonzero annihilator of M. It applies to every nonzero homogeneous ideal $I \subseteq S$, though $\operatorname{reg}(S/I)$ may have infinitely many corners.

For ordinary ideal powers in the standard graded setting, Swanson established linear bounds on regularity [99], and Cutkosky–Herzog–Trung and Kodiyalam proved eventual linearity [48, 81]. Combining our bound with a resolution of the Rees algebra gives the first multigraded analog of these results, showing that regularity of powers of ideals translates at a linear rate.

Corollary 1.5. [32, Thm. 5.1] *Let $I = \langle f_1, \ldots, f_s \rangle \subseteq S$ be a nonzero homogeneous ideal. There is* $\mathbf{a} \in \operatorname{Pic}(X)$ *such that for all integers $t > 0$ and all* $\mathbf{q}_1, \mathbf{q}_2 \in \operatorname{Pic}(X)$ *with* $\mathbf{q}_1 \geq \deg f_i \geq \mathbf{q}_2$ *for all i,*

$$t\mathbf{q}_1 + \mathbf{a} + \operatorname{reg}(S) \subseteq \operatorname{reg}(I^t) \subseteq t\mathbf{q}_2 + \operatorname{Nef}(X).$$

1.3 Top-Weight Cohomology of $\mathcal{A}_g$ The moduli stack of principally polarized abelian varieties of dimension g, $\mathcal{A}_g \coloneqq [\mathbb{H}_g/\mathrm{Sp}(2g,\mathbb{Z})]$, is one of the most classical objects in algebraic geometry, yet its rational cohomology is known completely only for $g \leq 4$ [76, 70, 74, 75], and until recently little was known beyond Borel's stable range [15]. In particular, Grushevsky asked whether $H^{2k+1}(\mathcal{A}_g;\mathbb{Q}) \neq 0$ for some g and k [68]. Chan, Galatius, and Payne's computation of the top-weight cohomology of $\mathcal{M}_g$, combining Deligne's weight filtration with the combinatorics of tropical moduli spaces, opened a powerful new approach to the cohomology of moduli spaces [40] that is now highly active [41, 94, 14, 37]. The PI and collaborators extended this program to $\mathcal{A}_g$. Since $\mathcal{A}_g$ lacks a modular compactification analogous to $\mathcal{M}_g \subset \overline{\mathcal{M}_g}$, this requires adapting [40] to toroidal compactifications whose boundary strata are indexed by cones in Voronoi's reduction theory. Analyzing these strata and relating the cohomology of $\mathcal{A}_g$ inductively to that of $\mathcal{A}_{g-1}$ and $GL_g(\mathbb{Z})$ yields many new classes.

Theorem 1.6. [20, Thm. A] *The rational cohomology satisfies $H^k(\mathcal{A}_g;\mathbb{Q}) \neq 0$ for* $(g,k) = (5,15)$, $(5,20)$, $(6,30)$, $(7,28)$, $(7,33)$, $(7,37)$, *and* $(7,42)$.

Theorem 1.6 answers Grushevsky's question affirmatively and, since $H^*(\mathcal{A}_g;\mathbb{Q}) \cong H^*(\mathrm{Sp}_{2g}(\mathbb{Z});\mathbb{Q})$, produces new unstable cohomology for the symplectic group. These results inspired a Hopf algebra structure on the top-weight cohomology of $\mathcal{A}_g$ [23] and Brown's bordification of $\mathcal{A}_g^{\mathrm{Trop}}$ [22], both producing further unstable cohomology of $\mathrm{GL}_n(\mathbb{Z})$. The framework of [20] is the template for the PI's subsequent work on tropicalizing general Shimura varieties [5], the subject of a plenary lecture at the 2025 AMS Summer Research Institute in Algebraic Geometry.

1.4 Broader Impacts from Prior NSF Support During the award, I organized five conferences: *WAGS* (100 participants), *Gems of Combinatorics I* (40), Spec($\overline{\mathbb{Q}}$) (50), *BATMOBILE 2022* (20), and *Gems of Commutative Algebra II* (40), plus special sessions at AMS Sectional Meetings. The *Gems of* series builds research networks for the next generation.

1.4.1 Mentoring & Virtual Mathematics. I led four undergraduate reading courses in algebra and geometry and volunteered with the *Berkeley Math Circle* to promote mathematics to high schoolers. In Summer 2021 I led 6 undergraduates on a research project related to [20], and in Summer 2022 I advised an undergraduate—now a math Ph.D. student with an NSF GRF—on a project on tropical abelian varieties. As mathematics moved online during COVID-19, I worked to make virtual activities reach those often at the periphery. In Summer and Fall 2020 I helped run Ravi Vakil's *Algebraic Geometry in the Time of Covid*, a massive open-access online course with

∼2,000 participants worldwide, and in Spring 2021 I organized an 8-week virtual undergraduate reading course in algebraic geometry and commutative algebra.

1.4.2 Community. During 2020–2023, I served on the board of *Spectra*, including as its inaugural president. I oversaw its growth and formalization, including the adoption of bylaws, the creation of an invited lecture at the JMM, and a fundraising campaign that has raised over $20,000.

2. The Geometry of Syzygies on Stacky Curves

Let X be a projective variety and L a very ample line bundle. For the embedding $X \hookrightarrow \mathbb{P}H^0(X,L) \cong \mathbb{P}^n$ defined by L, let $S_X = S/I_X$, where $S = \mathbb{C}[x_0,\dots,x_n]$ is the standard graded polynomial ring and I_X is the homogeneous defining ideal of X. The minimal graded free resolution of S_X

$$0 \longleftarrow S_X \xleftarrow{\epsilon} F_0 \xleftarrow{d_1} \cdots \qquad\qquad \cdots \xleftarrow{d_{k-1}} F_{k-1} \xleftarrow{d_k} F_k \longleftarrow \cdots$$

has terms $F_p = \bigoplus_q S(-q)^{\beta_{p,q}(X;L)}$, where $S(-q)_d = S_{d-q}$. The *graded Betti numbers (or syzygies)* $\beta_{p,q}(X;L)$ depend on the extrinsic embedding, not just on X; an overarching theme in the geometry of syzygies is understanding how they nonetheless capture the intrinsic geometry of X.

Forty years after Green's work, this connection is best understood for smooth non-hyperelliptic canonical curves of genus $g \geq 3$ ($L = K_X$). Green's conjecture [66] predicts the vanishing threshold of the quadratic strand:

$$\beta_{p,p+2}(X;K_X) = 0 \quad \Longleftrightarrow \quad p < \mathrm{Cliff}(X), \qquad 0 \leq p \leq g-3. \tag{1}$$

Here $\mathrm{Cliff}(X) \coloneqq \min\{\deg(L) - 2h^0(X,L) + 2 : h^0(X,L), h^1(X,L) \geq 2\}$, with the convention $\mathrm{Cliff}(X) = 1$ for a non-hyperelliptic curve of genus 3. Breakthrough work of Voisin proved this for general curves of every genus [103, 104]; new proofs [4, 78] and many refinements followed [59, 43, 57, 58, 52, 79].

This project develops the corresponding theory for *stacky curves* [2, 36, 1], where S has a nonstandard grading and the geometry appears only after accounting for fractional divisors. The classical theory of Noether, Petri, Mumford, and Green is a theory of standard graded rings, and it breaks down as soon as the divisor fails to be very ample. Already for a point O on an elliptic curve E, the section ring $\bigoplus_n H^0(E,nO)$ needs generators in degrees $1,2,3$, and its presentation is the Weierstrass equation $y^2 = x^3 + axz^4 + bz^6$, a sextic in $\mathbb{P}(1,2,3)$; the canonical ring of a hyperelliptic curve needs generators in degree 2, with the hyperelliptic equation a relation in degree 4. Stacky curves are the systematic setting for this phenomenon: by Demazure–Pinkham [96, 51, 105], every two-dimensional normal graded domain is $\bigoplus_n H^0(C, \lfloor nD \rfloor)$ for an ample $\mathbb{Q}$-divisor D on a curve C, i.e., a section ring on a stacky curve whose stacky points record the fractional part of D. The most important examples are modular curves: $X(1) = \mathcal{P}(4,6)$ has section ring $\mathbb{C}[E_4,E_6]$, and rings of modular forms on $X_0(N)$ and on Shimura curves are (log) canonical rings of stacky curves with stacky points at the elliptic points. Voight and Zureick-Brown [102] prove a stacky Noether–Petri theorem, but analogues of Mumford's normal generation, Green's N_p theorem, and Koszulness are open even for $X(1)$. The goal is a theory in which fractional divisors play the role that degree plays classically, extending Mumford's and Green's theorems and answering [27, Ques. 7.7].

2.1 Weighted Canonical Rings A stacky curve $\mathcal{X}$ is a smooth proper geometrically connected Deligne–Mumford stack of dimension one with a dense open subscheme. (Throughout this section, $\mathbb{K} = \mathbb{C}$.) Concretely, we may think of $\mathcal{X}$ as a curve C together with finitely many *stacky points* carrying cyclic stabilizers [102, Lem. 5.3.10], and the coarse space map $\pi\colon \mathcal{X} \to C$ forgets these stabilizers. If $x_1,\dots,x_s$ are the stacky points, $e_i = \#G(x_i)$, $P_i = \pi(x_i)$, and $g = g(C)$, the *signature* of $\mathcal{X}$ is $(g; e_1,\dots,e_s)$. Following [102], the *canonical divisor* on such a stacky curve is defined to be

$$K_{\mathcal{X}} = \pi^* K_C + \sum_i (e_i - 1)\, x_i = \pi^* D \qquad \text{where} \qquad D = K_C + \sum_i \left(1 - \tfrac{1}{e_i}\right) P_i \tag{2}$$

and its degree is $\deg(K_{\mathcal{X}}) = 2g - 2 + \sum_i (1 - 1/e_i)$, which may be a rational number. Thus the genus of $\mathcal{X}$, defined by $\deg(K_{\mathcal{X}}) = 2g_{\mathcal{X}} - 2$, is the rational number $g_{\mathcal{X}} = g + \frac{1}{2}\sum_i (1 - 1/e_i) \geq g$. We say $\mathcal{X}$ is *hyperbolic* when $\deg(K_{\mathcal{X}}) > 0$. The *canonical ring* of $\mathcal{X}$ is

$$R(\mathcal{X}, K_{\mathcal{X}}) \coloneqq \bigoplus_{d\in\mathbb{Z}} H^0(\mathcal{X}, dK_{\mathcal{X}}) \cong \bigoplus_{d\in\mathbb{Z}} H^0(C, \pi_*(dK_{\mathcal{X}})) \cong \bigoplus_{d\in\mathbb{Z}} H^0(C, \lfloor dD \rfloor),$$

where the *round-down* of a $\mathbb{Q}$-divisor $D = \sum_i a_i P_i$ on C is the honest divisor $\lfloor D \rfloor \coloneqq \sum_i \lfloor a_i \rfloor P_i$, and the last isomorphism follows from [102, § 5.4]. When $\mathcal{X}$ is understood, we write R for $R(\mathcal{X}, K_{\mathcal{X}})$.

This round-down is the essential difference from the classical theory. Since $\lfloor D \rfloor = K_C$, $R_1 = H^0(\mathcal{X}, K_{\mathcal{X}}) \cong H^0(C, K_C)$ has dimension $g(C)$ rather than $g(\mathcal{X})$. When stacky points are present, these sections vanish along the ramification divisor $K_{\mathcal{X}} - \pi^* K_C$, so they neither embed $\mathcal{X}$ nor generate R. Adding higher-degree generators yields an embedding into a weighted projective stack [19]; algebraically, R is a quotient of a polynomial ring with a nonstandard $\mathbb{Z}$-grading.

Following Brown and Erman [27, § 3.1], define the *minimal weighted canonical series* $\mathcal{W} = \bigoplus_{d\geq 1} \mathcal{W}_d$ by setting $\mathcal{W}_1 = R_1$ and, for $d \geq 2$, choosing $\mathcal{W}_d$ as a complement in R_d to the degree-d part of the subalgebra generated by $\mathcal{W}_1, \ldots, \mathcal{W}_{d-1}$. Then $\mathcal{W}$ embeds $\mathcal{X}$ into $\mathcal{P}(\mathcal{W})$, and $R \cong S/I_{\mathcal{X}}$, where $S = \mathrm{Sym}(\mathcal{W})$. For $g \geq 2$, Voight and Zureick-Brown's stacky Noether–Petri theorem gives sharp degree bounds of $e \coloneqq \max(1, e_1, \ldots, e_s)$ for generators and $2e$ for relations, subject to exceptional cases [102, Cor. 1.4.2]. These bounds do not specify the minimal generator degrees. The first goal of this project is to close this gap by making the generation bound exact.

Let $\nu_d = \dim \mathcal{W}_d$; the *weight multiset* of $\mathcal{W}$ has ν_d copies of each weight d, written d^{ν_d}. Set $N = \sum_d \nu_d$, so $\mathbb{P}(\mathcal{W}) = \mathbb{P}(1^{\nu_1}, 2^{\nu_2}, \ldots)$ has dimension $N - 1$, $\sigma = \sum_d d\,\nu_d$, and $c = N - 2$.

Conjecture 2.1. *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. The weight multiset of $\mathcal{W}$ is $(1^g,\ (2, 3, \ldots, e_1),\ \ldots,\ (2, 3, \ldots, e_s))$ with 2^{g-2} inserted if and only if C is hyperelliptic.*

This conjecture implies that the weight multiset already detects whether C is hyperelliptic, and that the generation degree is exactly e. We will prove this by producing a generator $y_{i,j} \in R_j$ of pole order exactly $j - 1$ at P_i for $2 \leq j \leq e_i$, and then showing these are minimal because $\lfloor d\frac{e-1}{e} \rfloor \leq d - 1$ forces a product of $k \geq 2$ generators of total degree j to have pole order at most $j - k$. That these elements in fact generate is the content of the following conjecture.

Filter R by the order of the poles at the stacky points by letting $F_\ell R_d \coloneqq H^0\big(C, dK_C + \sum_i \min\{\ell, b_i(d)\} P_i\big)$ with $b_i(d) = \lfloor d(e_i - 1)/e_i \rfloor$. This is a multiplicative exhaustive filtration on R, and $F_0 R = R(C, K_C)$. Let $T_e = \mathbb{C}[\Gamma_e]$ for the semigroup $\Gamma_e = \{(n, \ell) : 0 \leq \ell \leq \lfloor n\frac{e-1}{e} \rfloor\}$. The pole filtration gives a flat Rees degeneration from R to $\mathrm{gr}_F R$. We expect its special fiber to separate the principal parts at the stacky points, with products between distinct positive-pole contributions vanishing and coarse canonical sections acting by evaluation.

Conjecture 2.2. *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. Then $\mathrm{gr}_F R$ is the fiber product of $R(C, K_C)$ and $\prod_i T_{e_i}$ over $\prod_i \mathbb{C}[z_i]$, where the maps evaluate coarse sections at P_i and kill positive-pole monomials in T_{e_i}. In particular, there is an exact sequence*

$$0 \longrightarrow \mathrm{gr}_F R \longrightarrow R(C, K_C) \oplus \bigoplus_i T_{e_i} \longrightarrow \bigoplus_i \mathbb{C}[z_i] \longrightarrow 0. \tag{3}$$

Computations, including Figure 2, support this description. I will prove it by identifying the graded pieces and their multiplication through principal parts. With pole-order minimality, this would prove the formula in Conjecture 2.1. The resulting long exact sequence in Tor is the starting point for computing special-fiber syzygies and the cancellations recovering the Betti table of R.

2.2 The Shape of Stacky Green's Conjecture Classically, the Betti table of a canonical curve has four rows and $g - 1$ columns. The top and bottom rows each contain a single 1, so the content lies in the two middle rows, which are exchanged by the Gorenstein duality $\beta_{p,q}(C, K_C) = \beta_{g-2-p,\, g+1-q}(C, K_C)$. Thus Green's conjecture (1) does two things at once: (i) it determines the

coarse

	0	1	2	3	4
0	1	–	–	–	–
1	–	6	8	3	–
2	–	3	8	6	–
3	–	–	–	–	1

$(6;2)$

	0	1	2	3	4	5
0	1	–	–	–	–	–
1	–	6	8	3	–	–
2	–	8	24	24	8	–
3	–	–	3	8	6	–
4	–	–	–	–	–	1

$(6;3)$

	0	1	2	3	4	5	6
0	1	–	–	–	–	–	–
1	–	6	8	3	–	–	–
2	–	8	24	24	8	–	–
3	–	6	24	36	24	6	–
4	–	–	8	24	24	8	–
5	–	–	–	3	8	6	–
6	–	–	–	–	–	–	1

FIGURE 2. The Betti tables of a genus-6 trigonal canonical curve C and two stacky curves with signatures $(6;2)$ and $(6;3)$ and coarse space C.

length of the linear strand of every canonical curve, and (ii) together with the Hilbert function, it determines the entire Betti table of a general curve, since for general C the two middle rows do not overlap. Stacky canonical rings remain Gorenstein, but can have far more than four rows.

Theorem 2.3. *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. Let $R = S/I$ be a minimal graded presentation, where $S = \mathbb{K}[x_1, \ldots, x_N]$ has total weight σ, and let $c = N - 2$.*

(1) R is Gorenstein, with $\beta_{i,j}(R) = \beta_{c-i,\sigma+1-j}(R)$.

(2) The weighted regularity, in the sense of [27]*, of R is* 3*.*

(3) Its nonzero Betti entries lie in the row range $0 \leq j - i \leq \rho - 1$, where $\rho = \sigma - N + 4$; both endpoint rows are attained. Assuming Conjecture 2.1 for $g \geq 3$, we have that

$$\rho = \begin{cases} 4 + \sum_i \binom{e_i}{2}, & C \text{ non-hyperelliptic}, \\ g + 2 + \sum_i \binom{e_i}{2}, & C \text{ hyperelliptic}. \end{cases}$$

Intervening rows may be empty.

Parts (1) and (2) follow from a local cohomology calculation showing that R is a two-dimensional Cohen–Macaulay normal domain with $\operatorname{pd}_S R = c$, together with graded local duality, or alternatively from Watanabe's canonical module formula [105]. The pattern behind part (3) is striking: over a non-hyperelliptic coarse curve, whose canonical Betti table has 4 rows and $g - 1$ columns, each stacky point of order e adds $e - 1$ columns and $\binom{e}{2}$ rows. The two consequences of classical Green's conjecture thus become separate, new problems in the stacky world.

Research Problem 2.4. *Let $\mathcal{X}$ be a hyperbolic stacky curve with coarse space C, which is a smooth non-hyperelliptic curve of genus $g \geq 3$.*

(1) (Weak Stacky Green's Conjecture): Is the linear strand of $\beta(\mathcal{X}; K_{\mathcal{X}})$ determined by $\operatorname{Cliff}(C)$*?*

(2) (Strong Stacky Green's Conjecture): For a sufficiently general stacky curve $\mathcal{X}$, is $\beta(\mathcal{X}; K_{\mathcal{X}})$ determined by $\operatorname{Cliff}(C)$ *and its Hilbert function?*

In the following subsections, $\mathcal{X}$ and C are as in Research Problem 2.4, unless otherwise noted.

2.3 Towards a Weak Form of Stacky Green's Conjecture The weak form requires Green's N_p-conditions for nonstandard gradings. Brown and Erman call $Z \subset \mathbb{P}(\mathcal{W})$ weighted N_p if it is normally generated and F_i is generated in degree at most w^{i+1} for all $i \leq p$, where w^j is the sum of the j largest weights [27, Def. 1.2]; for standard weights, this is Green's classical N_p. Let $\tau_{\mathrm{BE}}(\mathcal{X})$ be the smallest $p \geq 1$ for which weighted N_p fails. For non-stacky curves, Brown and Erman prove a weighted N_p theorem: a smooth curve of genus g embedded by a normally generated weighted linear series $\mathcal{W}$ associated to a line bundle of degree at least $2g + 1$, with $\nu_1 > g$, satisfies weighted N_p for all $1 \leq p \leq \dim \mathcal{W} - g - 2$ [27, Thm. 1.7]. The stacky canonical series has $\nu_1 = g$, so their methods do not apply directly; their framework is nonetheless the right one, and [27, Ques. 7.7] asks for precisely the canonical analogue. We predict that the threshold is governed by the length of the coarse linear strand, $\ell(C) \coloneqq \max\{r \geq 1 : \beta_{r,r+1}(C; K_C) \neq 0\}$.

Conjecture 2.5 (General Weak Stacky Green's Conjecture)**.** *The threshold is* $\tau_{\mathrm{BE}}(\mathcal{X}) = (N-2) - \ell(C)$.

Green's conjecture predicts $\ell(C) = g - 2 - \mathrm{Cliff}(C)$, so $\tau_{\mathrm{BE}}(\mathcal{X}) = N - g + \mathrm{Cliff}(C)$; combined with Voisin's theorem [103, 104], Conjecture 2.5 would determine the threshold for a general coarse curve of every genus $g \geq 3$, uniformly over all signatures and positions of stacky points. The first ingredient in proving Conjecture 2.5 is the following comparison result for linear strands.

Proposition 2.6. *Assuming Conjecture 2.1, for every* $r \geq 1$*, the inclusion* $R(C, K_C) \hookrightarrow R$ *induces an isomorphism* $K_{r,1}(C, K_C) \xrightarrow{\sim} K_{r,1}(\mathcal{X}, K_{\mathcal{X}})$*. In particular,* $\beta_{r,r+1}(\mathcal{X}; K_{\mathcal{X}}) = \beta_{r,r+1}(C; K_C)$.

The proof of Proposition 2.6 is a direct Koszul calculation, using minimality of the weighted generators to identify linear-strand cycles and boundaries with those of the coarse canonical ring. Combined with Theorem 2.3, this gives the nonvanishing half of Conjecture 2.5. It remains to prove the vanishing half $\beta_{p,q}(R) = 0$ for $1 \leq p < c - \ell(C)$ and $q > w^{p+1}$. Equivalently, if u_k is the sum of the k smallest weights, duality asks for $\beta_{r,q}(R) = 0$ when $\ell(C) < r < c$ and $q \leq u_{r+1}$. For $r < g$ this follows already from the comparison and minimality; the new work is confined to $g \leq r < c$. I will study these remaining groups through the pole filtration and its Koszul spectral sequence given by Conjecture 2.2. The special fiber can have extra syzygies, so identifying its cancellation maps, rather than applying semicontinuity alone, is essential.

Finally, gonality and Clifford index supply no additional stacky invariant (here $\mathrm{Cliff}(\mathcal{X})$ is defined using line bundles with $h^0, h^1 \geq 2$, with the usual low-genus conventions).

Proposition 2.7 (Brill–Noether rigidity)**.** *We have* $\mathrm{Cliff}(\mathcal{X}) = \mathrm{Cliff}(C)$ *and* $\mathrm{gon}(\mathcal{X}) = \mathrm{gon}(C)$*. Every line bundle attaining either minimum is a pullback.*

Indeed, write $\mathcal{L} = \pi^* M \otimes \bigotimes_i \mathcal{T}_i^{a_i}$, where $\mathcal{T}_i = \mathcal{O}_{\mathcal{X}}(x_i)$ and $0 \leq a_i < e_i$. The round-down formula gives $\dim H^i(\mathcal{X}, \mathcal{L}) = \dim H^i(C, M)$ and $\deg_{\mathcal{X}} \mathcal{L} = \deg M + \sum_i \frac{a_i}{e_i}$. Nonzero fractional parts increase degree without changing cohomology. This proves both equalities and forces every minimizer to have trivial stacky characters. The remaining Betti rows may require finer invariants.

2.4 Toward a Strong Form of Stacky Green's Conjecture The strong form aims to recover the full Betti table from g, $\mathrm{Cliff}(C)$, and the signature; the PI has settled the simplest case.

Theorem 2.8 (One stacky point of order two)**.** *If* $\mathcal{X} = \sqrt[2]{(C, P)}$ *for* $P \in C$*, then for all* p, q*,*

$$\beta_{p,p+q}(R) = \beta_{p,p+q}\big(R(C, K_C)\big) + \beta_{p-1,p+q-2}\big(R(C, K_C)\big) + \delta_{q,2}\tbinom{g-1}{p} - \varepsilon_{p,q},$$

where $\varepsilon_{p,q} = 1$ *exactly for* $(p, q) \in \{(1,1), (0,2), (g-1,2), (g-2,3)\}$.

Thus Problem 2.4(2) has an affirmative answer here: $\beta(R)$ is independent of P. Indeed, $\mathcal{W}$ has weights $1^g, 2$, so $N = g+1$, $c = g-1$, and $\rho = 5$. Proposition 2.6 and Theorem 2.3 specify rows $0, 1, 3, 4$, leaving only row 2. In each internal degree j, the Hilbert numerator determines the alternating Betti sum and hence its sole unknown entry, $\beta_{j-2,j}$; by (2), the Hilbert series depends only on the signature. For $\rho \geq 7$, at least three rows remain unknown and these equations no longer suffice.

Conjecture 2.9 (Strong Stacky Green's Conjecture I)**.** *The Betti numbers* $\beta_{p,q}(\mathcal{X}; K_{\mathcal{X}})$ *depend only on* C *and the signature of* $\mathcal{X}$*, and not on the position of the stacky points.*

Since the Hilbert series depends only on the signature, Proposition 2.6 and Theorem 2.3 confine any positional dependence to the middle Betti rows, and Proposition 2.7 forces any invariant governing them to be local at the marked points rather than global on line bundles. This points toward pointed Brill–Noether theory with prescribed ramification, and toward limit linear series [**?**] and tropical methods for imposing ramification conditions [95, 85, 77, 45, 84].

The pole degeneration, together with the examples we have computed, suggests the following bold conjecture predicting the full Betti table of a canonical stacky curve. For a graded S-module

M, write $B_M(t,u) = \sum_{p,j} \beta^S_{p,j}(M) t^p u^j$. Let $J \subset S$ be the ideal generated by the variables of degree ≥ 2, and let P_C denote the Betti polynomial of the coarse canonical ring over S/J. Set $B_J^\vee(t,u) = t^{c-1}u^{\sigma+1}B_J(t^{-1},u^{-1})$.

Conjecture 2.10 (Strong Stacky Green's Conjecture II)**.** *We have*

$$B_{\mathcal{X}} = B_{S/J}\left(P_C + (1+tu)^{g-1}\sum_i \frac{u^2+\cdots+u^{e_i}}{1+tu^{e_i}}\right) - (1+t)(B_J + B_J^\vee).$$

The first term combines the coarse resolution with the higher-weight generators and the Koszul complexes on canonical forms vanishing at each P_i. The subtracted terms record consecutive cancellations paired by Gorenstein duality. I will use the fiber-product sequence (3) to identify the cancellations indexed by B_J in the special fiber, then show that deformation to R cancels precisely the dual pairs. To control these cancellations geometrically, M. Banks, P. Wei, and I have developed a weighted analogue of Lazarsfeld's kernel-bundle method [**?**, § 1]. Set $\mathcal{L} = \mathcal{O}_{\mathcal{X}}(K_{\mathcal{X}})$ and define

$$0 \longrightarrow \mathcal{M}_{\mathcal{W}} \longrightarrow \mathcal{E}_{\mathcal{W}} \coloneqq \bigoplus_{d\geq 1} \mathcal{W}_d \otimes \mathcal{L}^{-d} \longrightarrow \mathcal{O}_{\mathcal{X}} \longrightarrow 0.$$

Exterior powers and cohomology then identify the weighted syzygies:

$$\operatorname{Tor}_p^S(R,\mathbb{C})_j \cong \ker H^1\big(\wedge^{p+1}\mathcal{M}_{\mathcal{W}} \otimes \mathcal{L}^j \longrightarrow \wedge^{p+1}\mathcal{E}_{\mathcal{W}} \otimes \mathcal{L}^j\big)\,.$$

I will begin with $\mathcal{X} = \sqrt[3]{(C,P)}$, the first signature beyond the order-two case, with weights $1^g, 2, 3$. Writing $\mathcal{T} = \mathcal{O}_{\mathcal{X}}(x)$, successive evaluation maps filter $\mathcal{M}_{\mathcal{W}}$ with factors $\pi^* M_{K_C} \otimes \mathcal{L}^{-1}$, $\mathcal{L}^{-2} \otimes \mathcal{T}^{-1}$, and $\mathcal{L}^{-3} \otimes \mathcal{T}^{-1}$. Exterior powers and pushforward reduce the calculation to twists of $\wedge^a M_{K_C}$ by powers of K_C and multiples of P. The two extensions defining the filtration need not split; I will compute their connecting maps and combine them with the map to $\wedge^{p+1}\mathcal{E}_{\mathcal{W}}$ in the Tor formula above. For this signature, only rows $2, 3, 4$ remain undetermined, and duality exchanges rows 2 and 4, leaving $\beta_{p,p+2}$ and $\beta_{p,p+3}$ as the independent targets. I will calculate the relevant ranks for a general P, then determine whether they remain constant at special points. Establishing the predicted ranks would prove the full Betti formula in this first unresolved case. If ranks drop, the resulting determinantal loci would describe precisely where position independence fails and identify the pointed invariants needed to refine Conjecture 2.10.

2.5 Beyond Stacky Canonical Curves The proposed proof of Conjecture 2.2 never uses that the line bundle is $K_{\mathcal{X}}$, so we expect it to extend to any divisor on a stacky curve of the form $\mathcal{D} = \pi^* L \otimes \bigotimes_i \mathcal{T}_i^{a_i}$, where L is very ample on C and $\mathcal{T}_i = \mathcal{O}_{\mathcal{X}}(x_i)$ as above, so that $\deg \mathcal{T}_i = 1/e_i$ and $\mathcal{T}_i^{e_i} \cong \pi^*\mathcal{O}_C(P_i)$. Here $\ell(C)$ should be replaced by the Koszul cohomology invariant of $\lfloor \mathcal{D} \rfloor$, and this would answer [27, Ques. 7.6]. The PI is pursuing the complementary high-degree regime; adapting the weighted kernel-bundle method above, we aim to prove:

Conjecture 2.11. *Let $\mathcal{L} = \pi^* L \otimes \bigotimes_i \mathcal{T}_i^{a_i}$, with $0 < a_i < e_i$, $\gcd(a_i, e_i) = 1$, and $\deg L \geq 2g+1$. If $\mathcal{W}$ minimally generates $R(\mathcal{X},\mathcal{L})$, then $\tau_{\mathrm{BE}}(\mathcal{X},\mathcal{L}) = \tau(C,L) + \dim \mathcal{W} - h^0(C,L)$, where $\tau(C,L)$ is the first failure of ordinary N_p, and either threshold is ∞ if no failure occurs.*

The correction $\dim \mathcal{W} - h^0(C,L)$ counts the generators of weight greater than one: it is $\sum_i (e_i - 1)$ when every $a_i = e_i - 1$, but s when every $a_i = 1$. The proposed proof combines Watanabe's description of the canonical module [105] with Brown and Erman's bounds on strongly linear strands [27], reducing the first failure to a comparison with the coarse canonical module. The kernel-bundle approach will also address the remaining Betti numbers and extensions beyond this high-degree range. For log canonical rings $R(\mathcal{X}, K_{\mathcal{X}} + \Delta)$, the same cohomological framework applies even without the Gorenstein self-duality used above. This area is moving fast: beyond [27, 24, 7], Brown and Erman have studied positivity and nonstandard graded Betti numbers [25], Martinova has begun the asymptotic theory of syzygies of weighted projective spaces [89], and truncations of nonstandard graded polynomial rings are Koszul modules in the Herzog–Iyengar sense [50].

3. Multigraded Castelnuovo–Mumford Regularity: Bounds and Behavior

As discussed in Section 1.1, Castelnuovo–Mumford regularity links cohomological vanishing to the degrees of equations and syzygies. The PI's work extends this link to products of projective spaces and proves finite-generation and asymptotic results for regularity on toric varieties [32, 33]. Basic questions remain open: regularity can have infinitely many corners, it is not determined by the multigraded Betti numbers, and few geometric bounds are known. This project aims to explain the shape of the regularity region through the geometry of its support, bound its corners using syzygies, and bound the regularity of curves using intersection theory. Throughout, X, S, B, and $B_1, \dots, B_s$ are as in §1, with $n = \dim X$ and ground field $\mathbb{K}$ algebraically closed of characteristic zero.

3.1 The Shape and Minimal Degrees of Multigraded Regularity As the PI showed, the coordinate ring S/I_p of a point $p \in \mathcal{H}_2$ has infinitely many minimal regular degrees (Figure 1(c), [32]), although $\mathcal{O}_p$ is regular in every degree. These corners are measured relative to $\mathrm{Nef}(X)$, and I propose that their infinitude reflects the choice of cone: the ambient nef cone can be too small to account for the directions of positivity available to M. A cone that sees the module should recover a form of persistence and finite generation, up to a uniform shift. The point example supports this expectation: its support-relative regularity has only one corner. This leads to three connected problems: determine the directions of regularity from support geometry, bound its minimal degrees using syzygies, and control the growth of its corners for powers of ideals. For a finitely generated graded S-module M, write $\mathcal{F} = \widetilde{M} \neq 0$ and $Z = \mathrm{supp}(\mathcal{F})$. Writing D_ρ for the torus-invariant prime divisor associated to a ray ρ, set

$$K_Z = \bigcap_{p \in Z} \mathrm{cone}\{[D_\rho] : p \notin D_\rho\}, \qquad C_Z = \overline{\mathrm{cone}\{[C] : C \subset Z \text{ an integral curve}\}}^{\vee},$$

with curve classes in $N_1(X)_{\mathbb{R}}$. A rational class lies in K_Z when some positive multiple has an ambient linear system with no base point on Z, and in C_Z when its restriction to Z is nef; thus $\mathrm{Nef}(X)_{\mathbb{R}} \subset K_Z \subset C_Z$, where K_Z sees only which torus orbits meet Z and C_Z sees the Mori geometry of Z, toric or not. I conjecture that they are the recession cones of $\mathrm{reg}(M)$ and $\mathrm{reg}(\mathcal{F})$.

Conjecture 3.1. *If $H^0_B(M) = 0$, there are $\mathbf{a}_M, \mathbf{b}_M, \mathbf{a}_\mathcal{F}, \mathbf{b}_\mathcal{F} \in \mathrm{Pic}(X)$ such that*

$$\mathbf{a}_M + (K_Z \cap \mathrm{Pic}(X)) \subset \mathrm{reg}(M) \subset \mathbf{b}_M + K_Z \quad \textit{and} \quad \mathbf{a}_\mathcal{F} + (C_Z \cap \mathrm{Pic}(X)) \subset \mathrm{reg}(\mathcal{F}) \subset \mathbf{b}_\mathcal{F} + C_Z.$$

Module and sheaf regularity differ exactly by $H^1_B(M)$, the failure of sections to lift. In Figure 1(c), $\mathrm{reg}(S/I_p)$ fills $K_p \supsetneq \mathrm{Nef}(X)_{\mathbb{R}}$, while $C_p = \mathrm{Pic}(X)_{\mathbb{R}}$. For the diagonal $\Delta \subset \mathbb{P}^1 \times \mathbb{P}^1$, K_Δ is the nef cone but C_Δ is a half-plane: $\mathrm{reg}(S/(x_0y_1 - x_1y_0)) = \mathbb{N}^2$, while $\mathrm{reg}(\mathcal{O}_\Delta) = \{(a,b) : a + b \geq 0\}$, so one module corner shatters into infinitely many sheaf corners. I will approach the module bounds through toric modifications of linear systems whose base loci avoid Z. For sheaves, Fujita vanishing on Z [**?**, Thm. 1.5] and the global generation forced by regularity point to the two inclusions, and comparing first syzygies along support curves offers a route to effective translates [87].

For syzygies, our faithfulness theorem (§1) gives finitely many corners, but no box, and when $K_Z \supsetneq \mathrm{Nef}(X)_{\mathbb{R}}$, Conjecture 3.1 forces infinitely many corners. So we measure corners relative to $\Gamma_Z := K_Z \cap \mathrm{Pic}(X)$ instead. Define $\mathrm{reg}_Z(M)$ as Maclagan–Smith regularity with Γ_Z in place of $\mathrm{Nef}(X)$: $\mathbf{d} \in \mathrm{reg}_Z(M)$ if and only if $H^i_B(M)_{\mathbf{d} - \mathbf{g}_1 - \cdots - \mathbf{g}_{i-1} + \gamma} = 0$ for all $i \geq 1$ and $\gamma \in \Gamma_Z$, with each $\mathbf{g}_j$ among $B_1, \dots, B_s$ and the minimal generators of Γ_Z. Then $\mathrm{reg}_Z(M) \subseteq \mathrm{reg}(M)$ is Γ_Z-stable. It equals $\mathrm{reg}(M)$ on products, and in Figure 1(c), $\mathrm{reg}_Z(S/I_p) = \Gamma_p$ has one corner instead of infinitely many. I conjecture that, up to a uniform shift, regularity *is* support-relative regularity, generated in a box determined by the Betti table.

Conjecture 3.2. *Fix X and a basis of $\mathrm{Pic}(X)$. There are $\mathbf{c}_X \in \mathrm{Nef}(X)$ and, for each multigraded Betti table β, an explicit finite box $Q_X(\beta)$ such that every finitely generated graded S-module M with $\beta^S(M) = \beta$, $\widetilde{M} \neq 0$, and $H^0_B(M) = 0$ satisfies*

$$\mathbf{c}_X + \mathrm{reg}(M) \;\subseteq\; \mathrm{reg}_Z(M) \;=\; \big(Q_X(\beta) \cap \mathrm{reg}_Z(M)\big) + \Gamma_Z.$$

The inclusion is a support-relative nef persistence, and the equality says the Betti table locates every Γ_Z-corner; since Γ_Z contains a translate of $K_Z \cap \mathrm{Pic}(X)$, together they imply the module half of Conjecture 3.1, with $\mathbf{a}_M, \mathbf{b}_M$ depending only on β. For the inclusion, general forms of each degree in Γ_Z are nonzerodivisors on M, so the Maclagan–Smith argument [87] should run by induction on $\dim Z$: it gives equality when $\dim Z = 0$, and each further dimension should cost a bounded nef shift, uniform because Γ_Z depends only on which torus orbits meet Z. The box is the heart of the matter. On products, where $\mathbf{c}_X = \mathbf{0}$, ongoing work with Lauren Cranton Heller suggests $Q_X(\beta) = \prod_j [\ell_j(\beta), U_j(\beta)]$ with $\ell_j(\beta) = \min\{a_j : \beta_{0,\mathbf{a}} \neq 0\}$; we have proved this for $r = 2$ and are extending it via projections and partial Koszul complexes. In general, persistence along faces of Γ_Z and resolutions of the diagonal offer an approach [18, 72, 29].

Corollary 3.3. *Assume Conjecture 3.2, and fix X and β. Among modules satisfying its hypotheses, only finitely many regions $\mathrm{reg}_Z(M)$ occur. Each is determined by Γ_Z and the finite set $Q_X(\beta) \cap \mathrm{reg}_Z(M)$, and*

$$\mathrm{reg}_Z(M) \subseteq \mathrm{reg}(M) \subseteq \mathrm{reg}_Z(M) - \mathbf{c}_X.$$

There are only finitely many possible monoids Γ_Z, since K_Z depends only on which torus orbits meet Z. The corollary therefore reduces the possible regions to finitely many candidate degrees, but testing membership in $\mathrm{reg}_Z(M)$ still requires vanishing over all $\gamma \in \Gamma_Z$. I will seek a finite criterion for these tests, adapting the nonzerodivisor argument of [87, Cor. 4.8] to actual section degrees and using computable local cohomology [56]. Such a criterion would make the corollary effective. On products, $\Gamma_Z = \mathrm{Nef}(X)$, so the Maclagan–Smith criterion applies directly; alternatively, Theorem 1.3 computes regularity by quasilinear truncations. Combined with our box theorem, this gives a complete algorithm for $\mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$. Existing implementations, including ours [3], inspect a fixed window and can miss corners outside it. Our asymptotic theorem (§1) bounds $\mathrm{reg}(I^p)$ between translates moving linearly in p but does not exclude superlinear motion of corners along a face.

Conjecture 3.4. *Let $X = \mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_r}$, $I \subseteq S$ homogeneous, and M a finitely generated graded S-module. There are constants A, C such that for every corner $\mathbf{v} \in \mathrm{reg}(I^p M)$ we have $\|\mathbf{v}\|_\infty \leq Ap + C$.*

The Betti degrees of $I^p M$ are bounded linearly in p via the Rees module [**?**, Thm. 1.1], so this follows from Conjecture 3.2 if $U_j(\beta)$ is linear in the degrees of β. As corners form an antichain, the window allows $O(p^{r-1})$ of them, which is sharp: for $\mathfrak{m}$ generated by all Cox variables, $\mathrm{reg}(\mathfrak{m}^p) = \{\mathbf{d} \in \mathbb{N}^r : \sum_j d_j \geq p\}$ has $m(m-1)(R-1)/2 + m\epsilon$ corners.

3.2 A Gruson, Lazarsfeld, and Peskine Style Bound for Multigraded Regularity A foundational result of Gruson, Lazarsfeld, and Peskine [69] states that if $C \subset \mathbb{P}^r_{\mathbb{K}}$ is an integral nondegenerate curve defined over an algebraically closed field $\mathbb{K}$, then

$$\mathrm{reg}\,(I_C) \leq \deg(C) - r + 2,$$

where $I_C \subset \mathbb{K}[x_0, \ldots, x_r]$ is the (saturated) homogeneous defining ideal of C. Since $\deg(C) = C \cdot H$ for the hyperplane class H, the theorem is a strong bridge between the intersection theory of C and the degrees of its equations and syzygies. It sharpens classical work of Castelnuovo [38] on space curves and settles the curve case of the Eisenbud–Goto conjecture $\mathrm{reg}(I_C) \leq \deg(C) - \mathrm{codim}(C) + 1$ [55]. It has inspired decades of work on optimal bounds [97, 86, 82, 83] and on bounding regularity by the degrees of defining equations [12, 53, 49, **?**], and recent counterexamples to the general Eisenbud–Goto conjecture [90, 39, 91] make such geometric bounds all the more valuable.

Research Problem 3.5. *Prove a multigraded analogue of Gruson, Lazarsfeld, and Peskine's theorem for curves in smooth projective toric varieties.*

The objective is an explicit subset of $\mathrm{reg}(\mathcal{I}_C)$ whose shape reflects the divisor relations of the ambient variety [87]. Cobb has made progress toward Problem 3.5 by adapting the kernel

bundle techniques of Gruson, Lazarsfeld, and Peskine to products of projective spaces [44]. My proposed answer is modeled on the proof of the classical bound. Recall that $\mathcal{I}_C$ is m-regular if $H^q(\mathcal{I}_C(m-q)) = 0$ for $q \geq 1$; the sequence $0 \to \mathcal{I}_C \to \mathcal{O}_{\mathbb{P}^r} \to \mathcal{O}_C \to 0$ splits this vanishing into three questions. (i) *Ambient* ($q \geq 3$): since $H^{q-1}(\mathcal{O}_C(m-q)) = 0$, we have $H^q(\mathcal{I}_C(m-q)) \cong H^q(\mathcal{O}_{\mathbb{P}^r}(m-q))$, which vanishes at these twists because $\mathbb{P}^r$ has no intermediate cohomology. (ii) *Speciality* ($q = 2$): $H^2(\mathcal{I}_C(m-2)) \cong H^1(\mathcal{O}_C(m-2))$ vanishes when $\mathcal{O}_C(m-2)$ is nonspecial, e.g., when $(m-2)\deg(C) > 2p_a(C)-2$, which holds at $m = \deg(C)-r+2$ by Castelnuovo's genus bound. (iii) *Interpolation* ($q = 1$): $H^1(\mathcal{I}_C(m-1))$ is the cokernel of the restriction $H^0(\mathcal{O}_{\mathbb{P}^r}(m-1)) \to H^0(\mathcal{O}_C(m-1))$, so it vanishes if and only if every section on C extends to $\mathbb{P}^r$. This surjectivity, proved by kernel bundle arguments, is the heart of the theorem.

The constant $\deg(C) - r + 2$ also has a geometric meaning: by nondegeneracy, every secant line ℓ meets C in length at most $\deg(C) - r + 2$, and interpolation along ℓ shows that an s-secant line forces $\operatorname{reg}(\mathcal{I}_C) \geq s$. Thus the bound is governed by maximal secant lines.

Now let X be as in this section with $n \geq 2$. Say that an integral curve $C \subset X$ is *nondegenerate* if each restriction $H^0(X, B_j) \to H^0(C, B_j|_C)$ is injective. Set $r_j \coloneqq \dim_{\mathbb{K}} H^0(X, B_j) - 1$, $d_j \coloneqq \langle B_j \cdot C \rangle$, and the *defect* $\delta_j \coloneqq d_j - r_j$. Maclagan–Smith regularity of $\mathcal{I}_C$ at $\mathbf{d}$ tests the twists $\mathbf{d} - B_i$, then $\mathbf{d} - B_i - B_j$, and so on, so the same three questions arise. I propose answering each by a region depending only on the numerical class of C and on X:

(1) *Ambient vanishing.* Membership in $\operatorname{reg}(\mathcal{O}_X)$ gives $H^q(X, \mathcal{O}_X(\mathbf{d} - \sum_j v_j B_j)) = 0$ for $q > 0$ and $|\mathbf{v}| = q$. Since C is a curve, for $q \geq 3$ these are exactly the ideal-sheaf tests.

(2) *Nonspeciality.* Choose an explicit numerical bound $\widehat{G}_X([C]) \geq p_a(C)$. For example, $A = \sum_j B_j$ is very ample, and multiplication of the restricted section spaces shows that C spans at least a $\mathbb{P}^R$, where $R = \sum_j r_j$. Thus one may take Castelnuovo's bound $\pi(D, R)$, where $D = \sum_j d_j$: writing $D - 1 = m(R-1) + \epsilon$ with $0 \leq \epsilon < R - 1$, this is $\binom{m}{2}(R-1) + m\epsilon$. On a surface, adjunction gives the exact genus and can be used instead. Every second twist is nonspecial on

$$\mathcal{H}_X([C]) \coloneqq \left\{ \mathbf{d} \in \operatorname{Pic}(X) \;\middle|\; \mathbf{d} \cdot C \geq 2\widehat{G}_X([C]) - 1 + 2\max\{d_j\} \right\},$$

because $\deg \mathcal{O}_C(\mathbf{d} - B_i - B_j) > 2p_a(C) - 2$.

(3) *Interpolation.* Guided by the secant-line explanation above, set $M_X \coloneqq \overline{\operatorname{NE}}(X) \cap N_1(X)_{\mathbb{Z}}$ and $e_j(\gamma) \coloneqq B_j \cdot \gamma$. Call $0 \neq \gamma \in M_X$ *eligible* if $e_j(\gamma) < r_j$ for some j, and put

$$q_C(\gamma) \coloneqq \min_{j:\, e_j(\gamma) < r_j} \big(\delta_j + e_j(\gamma)\big), \qquad w(\gamma) \coloneqq \max_j e_j(\gamma).$$

If an integral rational curve $C' \neq C$ has eligible class γ, its image under B_j spans at most $\mathbb{P}^{e_j(\gamma)}$. The resulting sections vanishing on C', together with nondegeneracy of C, give $\operatorname{length}(C \cap C') \leq q_C(\gamma) + 1$. When C' is smooth, arbitrary values on a subscheme of length $q_C(\gamma) + 1$ can be interpolated by a line bundle of degree at least $q_C(\gamma)$. Requiring this degree at every first twist motivates:

$$\mathcal{S}_X([C]) \coloneqq \{\mathbf{d} \in \operatorname{Pic}(X) \mid \mathbf{d} \cdot \gamma \geq q_C(\gamma) + w(\gamma) \text{ for every eligible } \gamma \in M_X\}.$$

The definition tests all eligible lattice classes, including nonprimitive ones; the claim that these numerical inequalities suffice for interpolation is conjectural.

Conjecture 3.6. *If $C \subset X$ is an integral nondegenerate curve, then*

$$\mathcal{R}_X([C]) \coloneqq \mathcal{S}_X([C]) \cap \mathcal{H}_X([C]) \cap \operatorname{reg}(\mathcal{O}_X) \;\subseteq\; \operatorname{reg}(\mathcal{I}_C).$$

On $\mathbb{P}^r$, the line class gives $m \geq \deg(C) - r + 2$, higher eligible multiples give weaker inequalities, and Castelnuovo's genus bound supplies nonspeciality at the same threshold, so the conjecture recovers the classical theorem. The proposed bound improves on Cobb's even for a simple curve: the image $C \subset X = \mathbb{P}^2 \times \mathbb{P}^2$ of $[u : v] \mapsto \big([u^3 : u^2v : v^3], [u^2 : uv : v^2]\big)$, which pairs the normalization of a cuspidal plane cubic with the conic embedding and is therefore smooth. It has multidegree $(3, 2)$ and defects $(1, 0)$. The inequalities defining $\mathcal{S}_X([C])$ reduce to $a \geq 1$ and

$b \geq 2$; the genus estimate $\widehat{G}_X([C]) = \pi(5,4) = 1$ adds only the redundant $3a + 2b \geq 7$. In fact, $\mathcal{R}_X([C]) = \operatorname{reg}(\mathcal{I}_C) = (1,2) + \mathbb{N}^2$, so the bound is sharp (Lozovanu also obtains this corner over $\mathbb{C}$ [**?**, Thm. A]), whereas Cobb's bound gives $(3,3) + \mathbb{N}^2$ [44, Prop. B]. I have proved the conjecture for curves in $\mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$ with a minimal-degree projection, and computations for curves in $\mathrm{Bl}_p \mathbb{P}^3$ have found no counterexamples. What remains is precise: the ambient and nonspeciality conditions give the other regularity tests, so it suffices to prove that $H^0(X, \mathcal{O}_X(\mathbf{d} - B_i)) \to H^0(C, \mathcal{O}_C(\mathbf{d} - B_i))$ is surjective for every $\mathbf{d} \in \mathcal{R}_X([C])$ and each i. For general X, I will combine the classical auxiliary line bundle on the normalization with the Hanlon–Hicks–Lazarev resolution of the diagonal [69, 44, 72], whose terms are $\mathcal{D}_p = \bigoplus_\lambda \mathcal{O}_X(-A_{p\lambda}) \boxtimes \mathcal{O}_X(-E_{p\lambda})$. For the normalization $f\colon \widetilde{C} \to X$ and a line bundle L on $\widetilde{C}$, pullback, tensor product, and pushforward give a spectral sequence

$$E_1^{-p,q} = \bigoplus_\lambda H^q\Big(\widetilde{C}, L \otimes f^*\mathcal{O}_X(-A_{p\lambda})\Big) \otimes \mathcal{O}_X(-E_{p\lambda}), \qquad q = 0, 1,$$

converging to f_*L in total degree zero. The task is to choose L killing the relevant H^1 groups, then control the resulting presentation and its determinantal equations via the inequalities defining $\mathcal{R}_X([C])$; singular curves also require controlling the finite-length discrepancy from normalization. I will begin with Brown and Sayrafi's explicit resolution in Picard rank two [29], with $\mathrm{Bl}_p \mathbb{P}^3$ as a concrete test and curves in $\mathbb{P}^1 \times \mathbb{P}^2$ as a first graduate-student project. A proof would give intersection-theoretic bounds for equations in Cox rings [87] and, on products, for the quasilinear truncations of Theorem 1.3.

3.3 Multigraded Regularity of Torus-Invariant Subvarieties On a smooth projective toric variety $X = X_\Sigma$, the torus-invariant irreducible subvarieties are in dimension-reversing bijection with the cones of Σ. If $S = \mathbb{K}[x_\rho \mid \rho \in \Sigma(1)]$ is the Cox ring, the subvariety Z_σ corresponding to $\sigma \in \Sigma$ has Cox ideal $I_{Z_\sigma} = \langle x_\rho \mid \rho \in \sigma(1) \rangle$, which is resolved by a multigraded Koszul complex. Even so, the multigraded regularity of these subvarieties can be subtle: the four torus-fixed points of the Hirzebruch surface $\mathcal{H}_2$ exhibit two distinct regularities, one with infinitely many minimal corners [32]. These highly structured subvarieties are thus a natural testing ground for how regularity reflects positivity beyond syzygies, yet no concrete description of their regularity is known.

Research Problem 3.7. *Let $Z_\sigma \subset X$ be an irreducible torus-invariant subvariety corresponding to the cone σ. Give an explicit combinatorial characterization of* $\operatorname{reg}(I_{Z_\sigma})$ *and* $\operatorname{reg}(S/I_{Z_\sigma})$.

We ask for both because, unlike in the standard graded case, $\operatorname{reg}(S)$ need not have a single corner, so the two can behave quite differently [32]. Our approach combines the combinatorics of Hilbert functions with explicit computations of the higher cohomology of line bundles on torus-invariant subvarieties. With Jacob Lehman Duke, I have settled torus-fixed points. For a maximal cone σ, the variables surviving in S/I_{Z_σ} are indexed by rays outside σ, and the degrees of surviving monomials form the monoid $Q_\sigma = \{\sum_{\rho \notin \sigma(1)} a_\rho [D_\rho] \mid a_\rho \in \mathbb{N}\} \subset \operatorname{Pic}(X)$. By smoothness, the classes $[D_\rho]$ for $\rho \notin \sigma(1)$ form a lattice basis of $\operatorname{Pic}(X)$, so the Hilbert function of S/I_{Z_σ} is 1 on Q_σ and 0 elsewhere.

Proposition 3.8. *If $Z_\sigma \in X$ is a torus-fixed point corresponding to a maximal cone $\sigma \in \Sigma$, then*

$$\operatorname{reg}(S/I_{Z_\sigma}) = Q_\sigma \qquad \textit{and} \qquad \operatorname{reg}(I_{Z_\sigma}) = \operatorname{reg}(S) \cap \bigcap_{j=1}^{s} (B_j + Q_\sigma).$$

Since S/I_{Z_σ} has no higher sheaf cohomology, its regularity is the set of degrees where its Hilbert function equals its Hilbert polynomial [87, Prop. 6.4], namely Q_σ. A cohomology argument then reduces $\operatorname{reg}(I_{Z_\sigma})$ to surjectivity of evaluation at each first twist, i.e. $\mathbf{d} - B_j \in Q_\sigma$ for all j, giving the intersection. Notably, $\operatorname{reg}(S/I_{Z_\sigma})$ depends only on the rays of Σ and on σ, so it is unchanged across toric models sharing these data. For positive-dimensional torus-invariant subvarieties, a similar monomial count gives the Hilbert function of S/I_{Z_σ}, leaving the higher cohomology of line bundles

restricted to Z_σ. Since Z_σ is toric, with fan given by the images of the cones containing σ modulo σ, we plan to compute these vanishing regions inductively via toric Čech complexes.

Such descriptions would also make existing degeneration bounds explicit. Maclagan and Smith bound the regularity of a monomial quotient by intersecting shifted regularity regions of the factors in a Stanley filtration, which for saturated monomial ideals are coordinate rings of orbit closures [88, Thm. 4.1]. A combinatorial answer to Problem 3.7 would compute these directly from the fan, and the bounds pass from initial ideals to the quotient by semicontinuity [87, Prop. 6.13].

4. Uniform Homological Algebra: Regularity on the Cox Category

This part of the proposal is deliberately speculative: *multigraded regularity may be asking the right questions, but in the wrong category.* Work of the PI and others exposes the fragility of the classical link between cohomological vanishing and syzygies [87, 33, 32]. Even the Cox ring of the Hirzebruch surface $\mathcal{H}_2$, as a free module over itself, is not $\mathbf{0}$-regular [87, Ex. 4.3]. I propose to test whether treating birational models together provides a more natural setting.

Uniformity across toric birational models is a growing theme in multigraded algebra, beginning with Hanlon–Hicks–Lazarev's diagonal, which depends only on the fan rays [72]. Work on short resolutions, cellular constructions, and derived categories has developed this theme [26, 60, 61, 62, 47, 16, 10, 71]. Ballard et al. give a striking reformulation of King's conjecture: a full strong exceptional collection of line bundles can fail to exist on an individual smooth projective toric variety, yet exists in the *Cox category* incorporating its birational models [6]. I propose to use its uniform diagonal to "repair" classical regularity and make the structure sheaf $\mathbf{0}$-regular on every smooth projective model.

4.1 The Cox Category and Diagonal Heights Let X be a projective normal toric variety over an algebraically closed field $\mathbb{K}$ of characteristic zero, with Cox ring S. The maximal GKZ chambers give models X_i and smooth toric stacks $\mathcal{X}_i$, with $\mathcal{X}_i = X_i$ when X_i is smooth. Following Ballard et al., choose a common smooth toric stack with proper birational maps $\pi_i\colon \widetilde{\mathcal{X}} \to \mathcal{X}_i$ and set $D_{\mathrm{Cox}}(X) \coloneqq \langle \pi_i^* D^b(\mathcal{X}_i)\rangle \subseteq D^b(\widetilde{\mathcal{X}})$. Here $D^b(-) = D^b(\mathrm{Coh}(-))$, and all functors are derived. The category is independent of this choice and contains each $D^b(\mathcal{X}_i)$ fully faithfully. The divisor-class map $Q\colon \mathbb{Z}^{\Sigma(1)} \to \mathrm{Cl}(X)$ determines the half-open zonotope $Z_X = Q_{\mathbb{R}}((-1,0]^{\Sigma(1)})$ and the finite *Bondal–Thomsen collection* Θ of classes mapping into Z_X. Pulling back $\mathcal{O}(\alpha)$ from a chamber containing $-\alpha$ gives well-defined objects $T_\alpha \coloneqq \mathcal{O}_{\mathrm{Cox}}(\alpha)$ forming a full strong exceptional collection [6].

The uniform HHL complex lifts to a diagonal kernel inducing the identity on $D_{\mathrm{Cox}}(X)$ [6]. Take its minimal representative in the additive category generated by the fixed external products $T_\alpha \boxtimes T_\beta$, with terms $D_p = \bigoplus_{\alpha,\beta\in\Theta}(T_\alpha \boxtimes T_\beta)^{m_{p,\alpha,\beta}}$ for $p \geq 0$. Use the first factor as cohomological input, with $p = 0$ adjacent to the augmentation. For the occurring input labels $\alpha \in \Theta_H$, set $h(\alpha) \coloneqq \min\{p : m_{p,\alpha,\beta} \neq 0 \text{ for some } \beta\}$ and $\nu(\alpha) \coloneqq \max\{1, h(\alpha)\}$. The *height* records first occurrence in the minimal diagonal, not necessarily face codimension in the zonotope.

Definition 4.1 (Proposed Cox regularity)**.** *An object $\mathcal{E} \in D_{\mathrm{Cox}}(X)$ is* $\mathbf{0}$-regular *if*

$$\mathrm{Ext}^q(T_\alpha^\vee, \mathcal{E}) = 0 \qquad \text{for all } \alpha \in \Theta_H \text{ and } q \geq \nu(\alpha).$$

Degree-q cohomology in diagonal degree p contributes in position $p - q$. The cutoff leaves only positive positions at positive height and allows sections at height zero. On $\mathbb{P}^n$, $h(-j) = j$ for $0 \leq j \leq n$, recovering classical $\mathbf{0}$-regularity for coherent sheaves.

Research Problem 4.2. *Find a combinatorial formula for $h(\alpha)$ from the zonotope and GKZ fan.*

I will use Favero–Sapronov's stratified-torus methods and Han's minimal-resolution constructions [62, 71] to relate heights to Cox degrees and GKZ walls. In Figure 3, two labels of different heights become the same line bundle on the weighted projective stack.

4.2 The Structure Sheaf and Nef Geometry For a coherent sheaf $\mathcal{F}$ on a smooth projective model X_i, let $\mathcal{R}_i(\mathcal{F})$ be the classes $\mathbf{d} \in \mathrm{Pic}(X_i)$ for which $\pi_i^*(\mathcal{F}(\mathbf{d}))$ is $\mathbf{0}$-regular. The Θ-Transform Lemma [6, Lemma 1.6] and the projection formula give $\mathrm{Ext}^q(T_\alpha^\vee, \pi_i^*(\mathcal{F}(\mathbf{d}))) \cong H^q(X_i, \mathcal{F}(\mathbf{d} + \alpha))$.

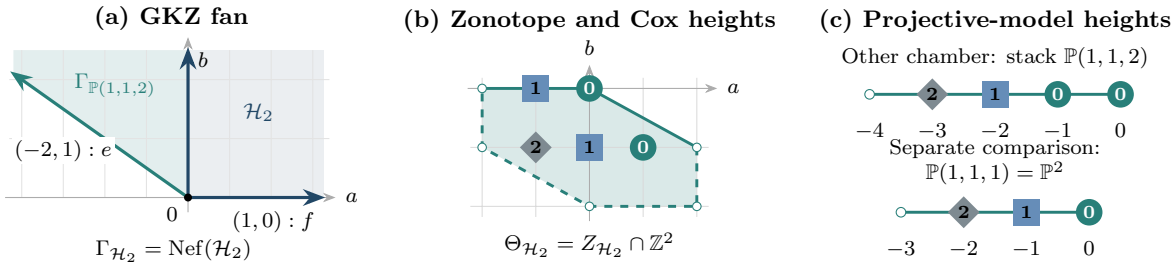


FIGURE 3. GKZ fan and diagonal heights for $\mathcal{H}_2$. (a) GKZ chambers; contracting e gives $\mathbb{P}(1,1,2)$. (b) Heights from the diagonal of [47, §4]; dashed edges and open vertices are excluded. (c) Heights on the stack $\mathbb{P}(1,1,2)$ and, for comparison, $\mathbb{P}^2$.

Lemma 4.3. *The object $\mathcal{O}_{\mathrm{Cox}}$ is $\mathbf{0}$-regular. On every smooth projective model X_i, every nef line bundle is $\mathbf{0}$-regular; in particular,* $\mathrm{Nef}(X_i)_{\mathbb{Z}} \subseteq \mathcal{R}_i(\mathcal{O}_{X_i})$.

Bondal–Thomsen vanishing gives $H^q(\mathcal{X}_i, \mathcal{O}_{\mathcal{X}_i}(\alpha)) = 0$ and $H^q(X_i, \mathcal{O}_{X_i}(\mathbf{d}+\alpha)) = 0$ for $q > 0$, $\alpha \in \Theta$, and nef $\mathbf{d}$ [6, Theorem 2.11]. The Θ-Transform Lemma and $\nu(\alpha) \geq 1$ imply both claims; the common pullback of the structure sheaves is $\mathcal{O}_{\mathrm{Cox}}$.

Question 4.4. *For every smooth projective model X_i and coherent sheaf $\mathcal{F}$, is $\mathcal{R}_i(\mathcal{F}) + \mathrm{Nef}(X_i)_{\mathbb{Z}} \subseteq \mathcal{R}_i(\mathcal{F})$? Is $\mathcal{F}(\mathbf{d})$ globally generated for every $\mathbf{d} \in \mathcal{R}_i(\mathcal{F})$?*

By Lemma 4.3, a positive answer would give $\mathcal{R}_i(\mathcal{O}_{X_i}) = \mathrm{Nef}(X_i)_{\mathbb{Z}}$, since globally generated line bundles are nef. Since nef translates of test line bundles can leave Z_X, I will seek resolutions of them by the original labels with controlled heights, starting with toric Koszul complexes on products and Hirzebruch surfaces, then scrolls and GKZ wall crossing. For global generation, I will analyze evaluation maps from degree-zero diagonal terms.

Every $\mathcal{E} \in D_{\mathrm{Cox}}(X)$ is represented by a bounded minimal complex of finite direct sums of the T_β, unique up to isomorphism of complexes. Write $C_p(\mathcal{E}) = \bigoplus_{\beta \in \Theta} T_\beta^{b_{p,\beta}(\mathcal{E})}$ for $p \in \mathbb{Z}$. The multiplicities $b_{p,\beta}(\mathcal{E})$ are its *Cox Betti numbers*.

Question 4.5. *What is the relationship between the proposed regularity and Cox Betti numbers?*

The uniform diagonal constructs a Cox complex from the cohomology defining regularity; minimization determines which terms survive. Guided by classical regularity, I will study how this passage relates cohomological vanishing to degrees and homological positions in minimal Cox complexes.

5. Extending Computational Algebra Software

Computation drives much of the recent progress in multigraded commutative algebra and toric geometry, and the PI has a track record of building such tools. Each project below is targeted and produces a concrete community tool that also supports this proposal.

5.1 The `toricStacks` Package for *Macaulay2* The lack of software is a major bottleneck for homological invariants of stacks. A general framework is out of reach, but toric stacks are tractable: they correspond to *stacky fans*, whose well-developed combinatorics makes computation feasible [63, 64, 17], but no implementation exists. Given the success of *NormalToricVarieties* [**?**], a *toricStacks* package would benefit the community and directly support the experiments in §2.

Research Problem 5.1. *Develop a Macaulay2* [**?**] *package for computations with toric stacks.*

The package will (i) construct toric stacks from stacky fans and from families such as weighted projective stacks and root stacks, (ii) compute the Cox ring with its $\mathrm{Cl}(\mathcal{X})$-grading and irrelevant ideal, and (iii) support coherent sheaves, including sheaf cohomology via local cohomology [56]. This is joint work in progress with Maya Banks and John Cobb; (i) is complete and (ii) is underway.

5.2 A Database of Human-Generated Examples in Algebra Good examples drive conjectures, theorems, and algorithms, and a trove of human-generated examples sits unused in the documentation and tests of open-source computer algebra systems. *Macaulay2* and its roughly 250 packages contain thousands of rings, ideals, and modules, nearly all chosen by experts to illustrate something interesting. Yet they are scattered across source files, not easily accessible.

Research Problem 5.2. *Build a Macaulay2 package collecting the algebraic objects in its source and documentation into a queryable database with provenance and invariants.*

The package will parse the `EXAMPLE` and `TEST` blocks of installed packages, re-execute them in a sandbox, and record each resulting object as directly callable, with its source location and authorship. Much of the parsing already exists; the work lies in execution and curation. The result would be a testbed for conjectures and a benchmark suite easing future *Macaulay2* development. As calls grow for AI and machine learning tools built by mathematicians on mathematician-owned data, this is a small first step, with all authorship tracked and acknowledged.

6. Broader Impacts

6.1 Organizing. Building on Section 1.4, since 2022 I have organized *Gems of Combinatorics II* (30 participants), *AGNES* (150), and *BATMOBILE 2024 & 2025* (50/30), and I co-organize Dartmouth's *Algebra & Number Theory* seminar. *Future Plans:* With co-organizers, I have proposed a *Thematic Semester in Combinatorial Algebraic Geometry* to the Fields Institute for 2030. I will host the next *Gems of Commutative Algebra* at Dartmouth in Fall 2028 and write a *Notices of the AMS* article on organizing early-career conferences. John Voight and I are in the planning stages of a Macaulay2+Magma workshop on the future of mathematical software.

6.2 Open Educational Resources. In Fall 2026 I am lecturing for the Preliminary Arizona Winter School, a nine-week virtual course with 300 students (200 undergraduate and master's students); all material (videos, 100 pages of IBL notes and problems, etc.) is freely available. Building on work of Karen Smith and Jack Jeffries, I designed an inquiry-based commutative algebra course of 16 worksheets (200+ pages). Both are on my website. *Future Plans:* I will develop these materials into an open-access introduction to algebraic geometry with Macaulay2 exercises.

6.3 Open Computational Resources. I created *Syzygy Data*, `syzygydata.com`, a public database of large syzygy computations, and co-authored four Macaulay2 packages. *Future Plans:* I will lead a group developing `toricStacks` at the 2027 *Macaulay2* conference at Auburn.

6.4 Campus Community. I am the faculty advisor to Dartmouth's *AWM Student Chapter* and work with the campus *oSTEM* chapter to broaden participation in STEM. In Fall 2025 I founded *Unwritten Rules*, a graduate seminar that makes the hidden curriculum of an academic career explicit. *Future Plans:* I will continue *Unwritten Rules* and extend it to Dartmouth postdocs.

6.5 National & International Service. I have worked with the *AWM* Executive Committee to expand support for a broader range of mathematicians. From Winter 2023 until its dissolution in Spring 2025, I served on SLMath's *Committee on Women in Mathematics*. I now serve on the *AMS Committee on Engagement, Participation, & Advancement* and the *AMS Simons Travel Grant Committee*. In January 2026 I began a three-year term on the Board of Trustees of *Spectra*.

6.6 Mentoring. I advise three Ph.D. students on problems central to this proposal: Will D. (algebraic geometry + AI/ML), Jacob L-D. (regularity of torus-invariant subvarieties), and Alex M. (positivity of the BT collection). At MREG 2023 I began mentoring graduate students Bailee Z. (Michigan) and Jacob B. (KSU) on a project that led to [30]. In Summer 2026 I mentored three undergraduates through *Count Me In* (organized by M. Hering and D. Maclagan), and since Fall 2024 I have mentored two Dartmouth undergraduates. *Future Plans:* I will organize a summer undergraduate research program through Dartmouth's SHUR program.

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